Concentration Spectrum of the Half-Filled Sine Kernel: Periods, Tails and Plunge Windows

Kirtus Justus

Theoretical Physicist

Port Coquitlam, British Columbia, Canada

Baseline edition, 12 November 2026

Scope

Five articles on the discrete concentration problem for the half-filled sine kernel: higher-order modular periods, concentration eigenvalues from logarithmic connection coefficients, spectral tails and exact concentration logits, reciprocal traces and the plunge law, and positive moment bounds on growing plunge windows. They supply reference-kernel inputs used in the Wilson–Dirac relative-entropy papers.

Contents

Part	Title	Page
1	Higher-Order Modular Periods for the Half-Filled Sine Kernel	3
2	Concentration Eigenvalues from Logarithmic Connection Coefficients	11
3	Spectral Tails and Exact Concentration Logits	16
4	Reciprocal Traces and the Half-Filled Plunge Law	23
5	Positive Moment Bounds and Growing Plunge Windows	30

Formal statements

Identifiers are those of the program theorem registry. Page numbers refer to this file.

Identifier	Statement	Page
D6-01-01	Proposition 1 (Finite Fourier equation)	4
D6-01-02	Theorem 2 (Formal coefficient identities)	5
D6-01-03	Theorem 3 (Edge limits of the formal functions)	6
D6-01-04	Proposition 4 (Exact-arithmetic quadrature bound)	7
D6-02-01	Lemma 1 (Exact Fourier equation and commutation)	12
D6-02-02	Lemma 2 (Wronskian and norming identities)	12
D6-02-03	Theorem 3 (Complementary-band connection formula)	13
D6-02-04	Corollary 4 (Removal of the finite characteristic polynomial)	13
D6-02-05	Theorem 5 (Relative-error criteria)	14
D6-03-01	Lemma 1 (Compact spectrum and elementary bounds)	17
D6-03-02	Theorem 2 (Exact selection of the first Nmodes)	17
D6-03-03	Lemma 3 (Trace identity without a normalization ambiguity)	18
D6-03-04	Theorem 4 (Spectral-tail formula)	19
D6-03-05	Corollary 5 (Explicit omitted-spectrum bound)	19
D6-03-06	Corollary 6 (Acceleration and a quantitative certificate)	20

Identifier	Statement	Page
D6-04-01	Lemma 1 (Weighted Legendre trace)	24
D6-04-02	Lemma 2 (Mellin reflection at half-integers)	25
D6-04-03	Theorem 3 (Positive Laplace formula)	25
D6-04-04	Corollary 4 (An exact slope bound)	26
D6-04-05	Theorem 5 (Asymptotic expansion with a remainder)	26
D6-04-06	Theorem 6 (Signed plunge law for shrinking Jacobi modes)	27
D6-04-07	Lemma 7 (An elementary central-mode bound)	27
D6-05-01	Lemma 1 (Autocorrelation identity)	31
D6-05-02	Theorem 2 (Stieltjes representation and signed remainder)	31
D6-05-03	Corollary 3 (Two rational slope bounds)	32
D6-05-04	Theorem 4 (Explicit logarithmic-window bound)	32
D6-05-05	Lemma 5 (Explicit ratio control)	34
D6-05-06	Corollary 6 (Uniform signed plunge law on a growing window)	34
D6-05-07	Theorem 7 (Exact deletion of the finite block)	35
D6-05-08	Theorem 8 (Higher-moment leading terms with an explicit error)	35
D6-05-09	Theorem 9 (Uniform center law beyond shrinking eigenvalues)	36
D6-05-10	Corollary 10 (A logarithmic number of actual central pairs)	37
D6-05-11	Theorem 1.3	38

Acknowledgments and contributions

Kirtus Justus directed the research questions, the selection of mathematical models and the publication standards, is the sole named author, and is responsible for the claims presented here. ChatGPT (OpenAI) assisted with derivations, critical checks and manuscript preparation. Claude (Anthropic), Gemini (Google), Grok (SpaceXAI) and DeepSeek (High-Flyer) contributed to the wider research process; attribution of particular arguments is limited to traceable records. Editorial preparation of this edition: Claude (Anthropic). Assistance does not imply provider endorsement, external peer review or institutional endorsement.